

Precise Definitions and Counterexamples

Two jobs a definition has to do. A precise definition must (1) be true of *every* object it names and (2) be true of *nothing else*. A statement that fails job (2) is not a definition — it is at best a property, and the way you prove that is to exhibit one object that satisfies the statement and is not the thing: a **counterexample**. One counterexample is enough.

Model: definitions become coordinate tests. Every definition below can be checked with arithmetic, which is what makes a counterexample something you can *compute* rather than argue about.

Word	Precise definition	Coordinate test (notation)
midpoint of $\overline{AB}$	the point M on $\overline{AB}$ with $AM = MB$	$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
congruent segments	segments of equal length	$AB = CD$, each by $\sqrt{(\Delta x)^2 + (\Delta y)^2}$
parallel lines	coplanar lines with no point in common	slopes equal: $m_1 = m_2$
perpendicular lines	lines meeting at a right angle	$m_1 \cdot m_2 = -1$

Directions. Use the definition, not the picture in your head. Show the arithmetic that settles each question, and write your final answer on the answer line.

1. A student writes: “The midpoint of a segment is a point near the middle of it.”

Use the *precise* definition — the point M on $\overline{AB}$ with $AM = MB$ — to find the midpoint of $\overline{AB}$ where $A(-6, 2)$ and $B(4, 8)$. Then state in one sentence why the student’s wording fails to name one particular point.

Answer: _____

2. By definition, a triangle is *isosceles* when at least two of its sides are congruent. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, $R(3, 4)$, and $PQ = 6$.

Compute PR . Then decide whether $\triangle PQR$ is isosceles, and say which wording — “at least two congruent sides” or “exactly two congruent sides” — decides whether an equilateral triangle counts as isosceles.

Answer: _____

3. Kai writes: “The perpendicular bisector of $\overline{AB}$ is the line through the midpoint of $\overline{AB}$.” Let $A(-4, 1)$ and $B(2, 5)$, so the midpoint is $M(-1, 3)$ and the slope of $\overline{AB}$ is $\frac{2}{3}$. Kai draws $\overleftrightarrow{CM}$ through $C(3, -2)$ and M .

Compute the slope of $\overleftrightarrow{CM}$. Is the product of the two slopes -1 ? Explain what your answer proves about Kai’s statement.

Answer: _____

4. Consider the true statement: “If a quadrilateral is a square, then it has four right angles.”

(a) Write its converse. (b) Rectangle $PQRS$ has $P(0, 0)$, $Q(6, 0)$, $R(6, 3)$, $S(0, 3)$; all four of its angles are right angles and $PQ = 6$. Compute QR . (c) Explain why $PQRS$ makes the converse false.

Answer: _____

5. Consider the claim: “If two segments have the same length, then they are parallel.”

$\overline{AB}$ runs from $A(0, 0)$ to $B(3, 4)$ and $\overline{CD}$ runs from $C(0, 0)$ to $D(-4, 3)$. Both have length 5, and the slope of $\overline{AB}$ is $\frac{4}{3}$.

Compute the slope of $\overline{CD}$, then explain in one sentence why these two segments form a counterexample.

Answer: _____

6. By definition, two non-vertical lines are perpendicular exactly when the product of their slopes is -1 .

Line 1 passes through $(0, 0)$ and $(4, 3)$, so its slope is $\frac{3}{4}$. Line 2 passes through $(0, 0)$ and $(3, m)$.

Find the value of m that makes Line 2 perpendicular to Line 1. Set up the defining equation before you solve it.

Answer: _____

7. A student writes: “A rhombus is a quadrilateral whose diagonals are perpendicular.”

Quadrilateral $WXYZ$ has $W(0, 0)$, $X(3, 4)$, $Y(0, 10)$, $Z(-3, 4)$. Its diagonal $\overline{WY}$ is vertical and its diagonal $\overline{XZ}$ is horizontal, so the diagonals are perpendicular. Its side lengths are $WX = 5$, $XY = \sqrt{45}$, $YZ = \sqrt{45}$, $ZW = 5$.

(a) Explain why $WXYZ$ satisfies the student’s statement but is not a rhombus. (b) Write a precise definition of *rhombus*. (c) The student’s statement is still true of every rhombus. What is it, if not a definition?

8. Decide whether this claim is true: “If the diagonals of a quadrilateral are congruent, then the quadrilateral is a rectangle.”

If it is false, plot a labelled counterexample on the grid, list its vertices, compute both diagonal lengths to show they are congruent, and name the part of the definition of *rectangle* that your figure fails.

